PreSareeFinder – Budget-Based Saree Discovery Platform for Hyderabad

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Abstract

PreSareeFinder is a platform that helps users explore saree collections from Hyderabad stores before visiting shopping malls. Customers can browse sarees based on budget and style such as silk, cotton, or designer sarees, save favourites, add items to cart, and view billing details. Store admins can add new sarees, update prices, and manage stock. The project combines Front-End Web Development and Data Structures & Algorithms, using HTML, CSS, JavaScript, and local Storage for the interface and data handling, while searching, sorting, and data structures are used to efficiently manage saree information.

Existing System

- There is no single platform to compare sarees across multiple Hyderabad stores.
- Customers must visit each store physically to check saree designs, prices, and availability.
- Store owners mostly manage inventory manually without any digital system.
- There is no proper separation between customer browsing and store admin management.
- Existing e-commerce platforms are complex and require backend servers, which small stores cannot easily maintain.
- Traditional systems do not use efficient data structures or algorithms to organize, search, or manage saree information.
- Small local Hyderabad stores lack a simple online presence to showcase their collections.

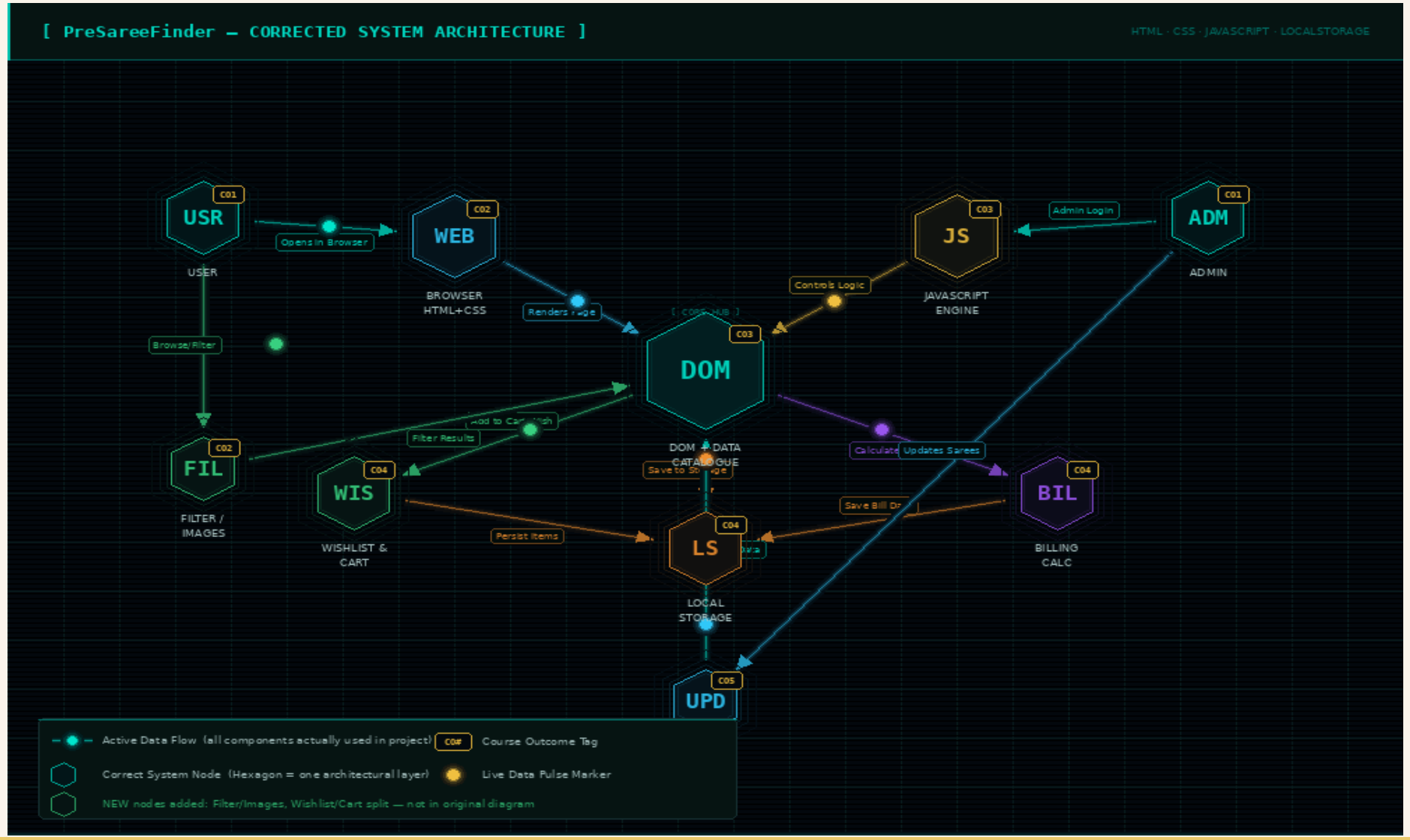
Proposed System

- A single platform that displays sarees from multiple Hyderabad stores in one place.
- Role-based login system providing separate access for customers and store admins.
- Customers can search sarees by name and filter by price or category using efficient searching methods.
- Users can save sarees to Wishlist and add items to Cart for easy selection.
- Store admins can add, edit, delete sarees and update stock details efficiently.
- Data is managed using local Storage and organized using data structures and algorithms for faster retrieval.
- The system provides a simple and responsive interface that works smoothly on desktop browsers.

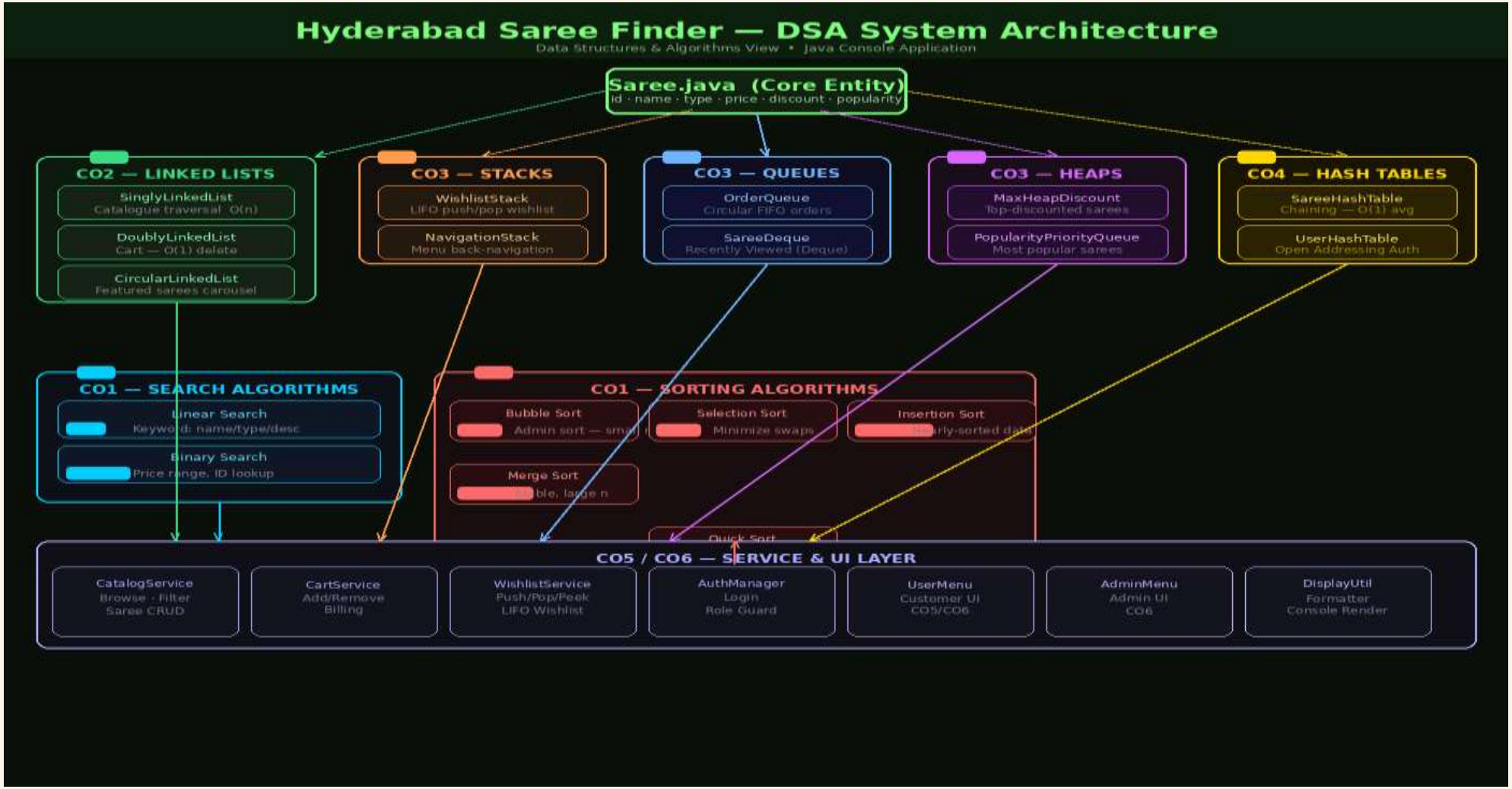
System Architecture

- Every page first loads saree data from browser storage or data structures when it opens.
- The system then checks the user role (Customer or Admin) after login.
- If an unauthorized user tries to access a page, they are redirected back to the login page.
- Customers can view saree cards, search, and browse collections, while admins see their store's inventory management panel.
- Any change made by an admin (add, update, delete sarees) is immediately updated in the system data.
- Customers always see the latest saree information when they open or refresh the page.
- Cart and Wishlist data are saved, so selected items remain available even after refreshing or reopening the application.

FWD System Architecture



DSA System Architecture



Project Modules

User Authentication Module – Handles login and role-based access for customers and admins.

Saree Catalog Module – Displays saree collections with details like name, price, type, and store.

Search and Filter Module – Allows users to search sarees and filter by price or category.

Wishlist Module – Enables customers to save favourite sarees for later.

Cart Management Module – Allows users to add or remove sarees from the cart.

Billing Module – Calculates the total price of selected sarees.

Admin Management Module – Allows admins to add, update, or delete saree details.

Data Management Module – Stores and manages saree data (LocalStorage in FWD / Data Structures in DSA).

Objectives

- Develop a single website that displays sarees from multiple Hyderabad stores in one place.
- Provide separate login access for customers and admins.
- Enable customers to search and filter sarees by name, price, and category.
- Allow admins to manage store inventory by adding, editing, deleting sarees, and updating stock.
- Maintain cart and wish-list data so it remains available even after refreshing the page.
- Build a desktop-compatible web application that runs directly in the browser without installation.

Co’s Attainments - FWD

CO1: Developed a multi-page web application using HTML5, CSS3 and JavaScript

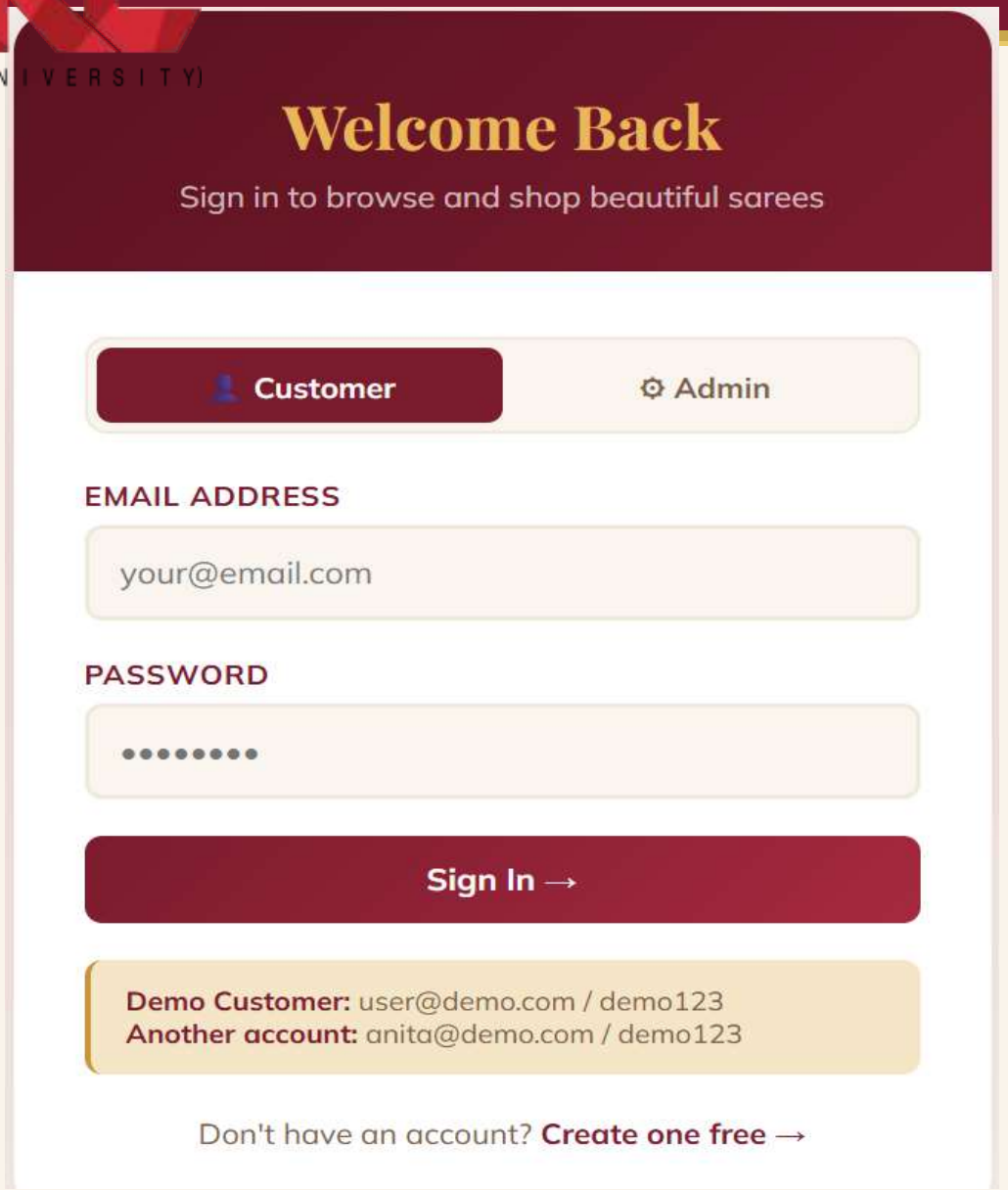
CO2: Implemented role-based login system for customers and store admins

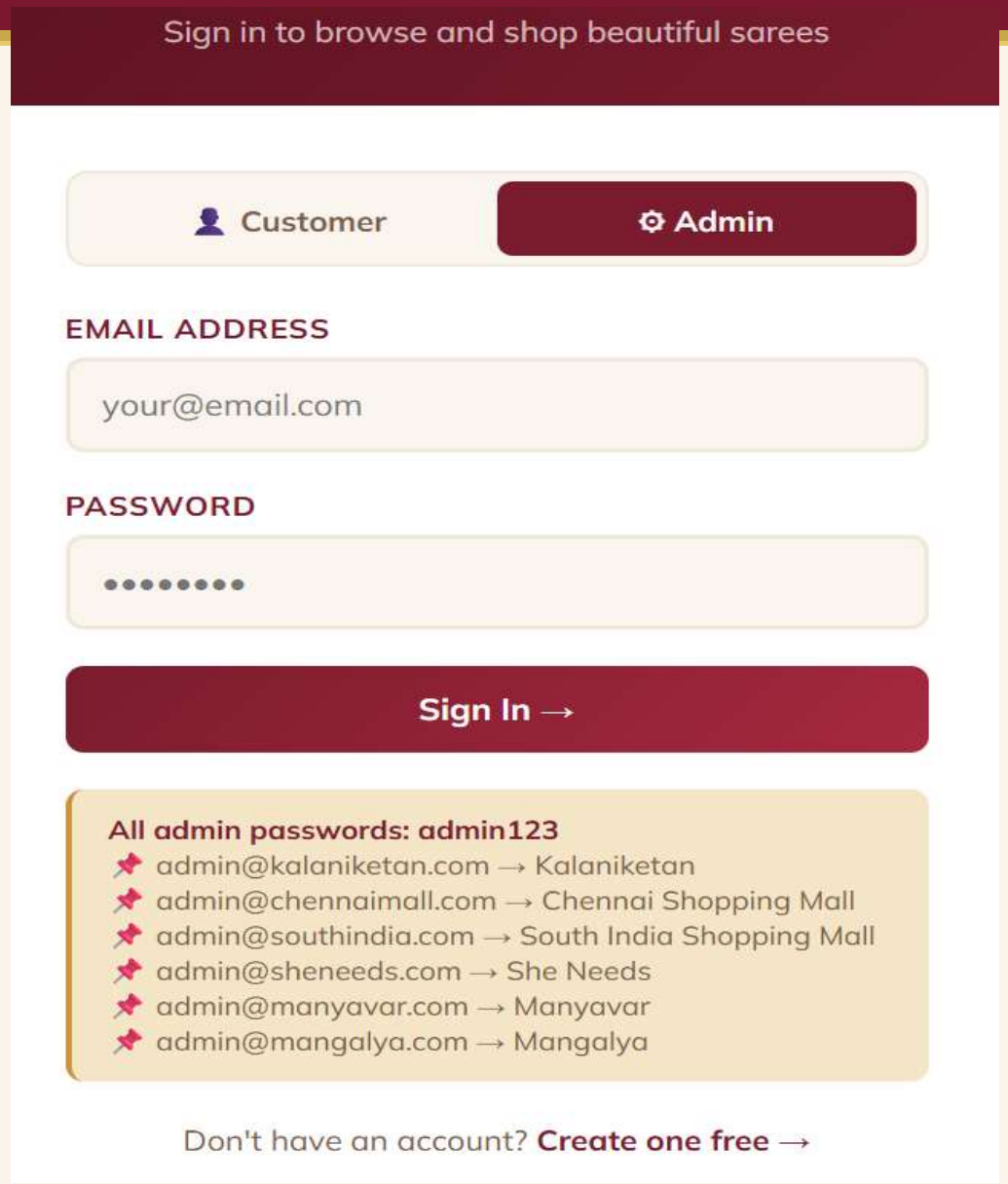
CO3: Built saree browsing with search, filter and modal detail view

CO4: Implemented wish-list and cart using local Storage for persistence

CO5: Implemented admin CRUD operations with store-specific access control

Login Page


Welcome Back
Sign in to browse and shop beautiful sarees
Customer
Admin
EMAIL ADDRESS
your@email.com
PASSWORD
Sign In →
Demo Customer: user@demo.com / demo123
Another account: anita@demo.com / demo123
Don't have an account? Create one free →


Sign in to browse and shop beautiful sarees
Customer
Admin
EMAIL ADDRESS
your@email.com
PASSWORD
Sign In →
All admin passwords: admin123
admin@kalaniketan.com → Kalaniketan
admin@chennaimall.com → Chennai Shopping Mall
admin@southindia.com → South India Shopping Mall
admin@sheneeds.com → She Needs
admin@manyavar.com → Manyavar
admin@mangalya.com → Mangalya
Don't have an account? Create one free →

Welcome Page

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Saree Finder

Home | Sarees | Wishlist 3 | Cart 2 | Hello, Priya | Logout

Welcome back, Priya! 👋

Explore the finest sarees from across Hyderabad

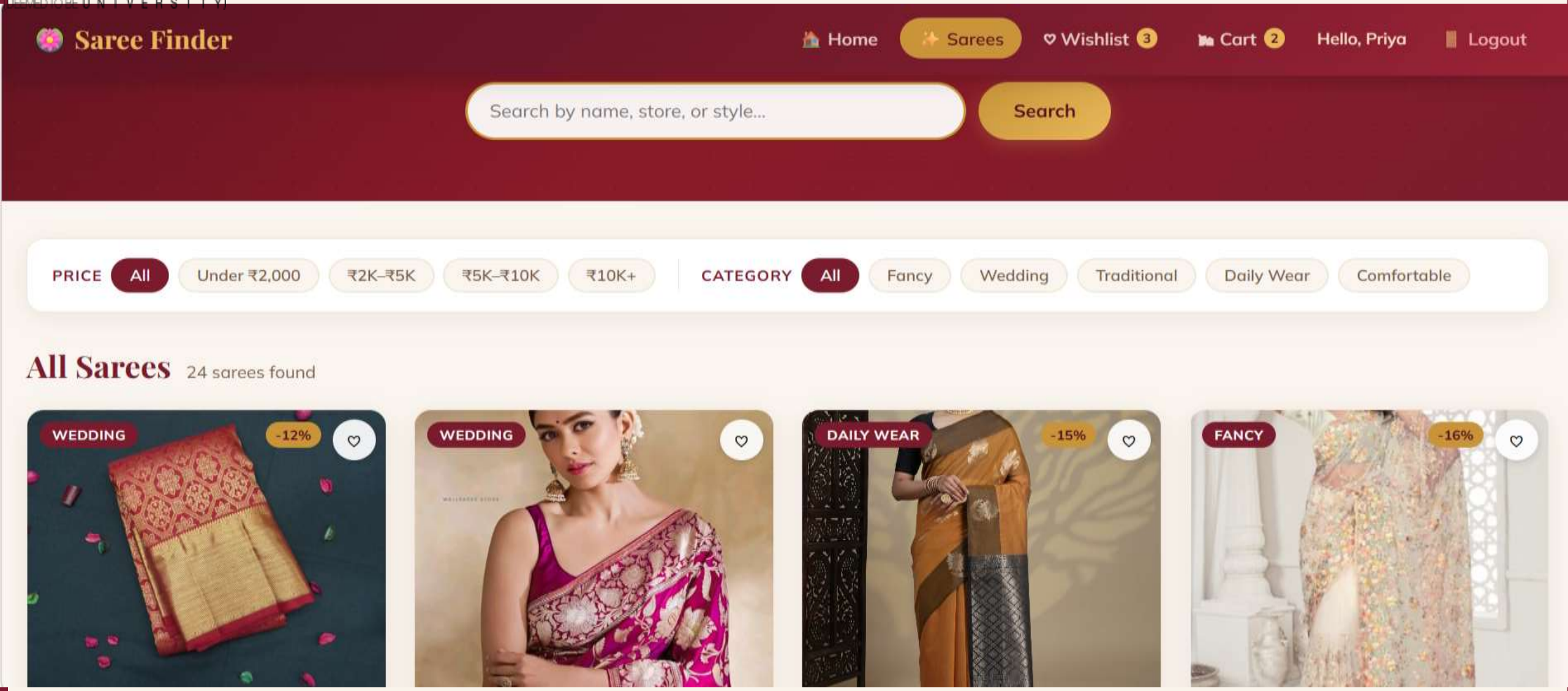
What would you like to do?

- Browse Sarees — Explore all 20+ sarees
- My Wishlist — 3 saved items
- My Cart — 2 items
- Wedding Sarees — Bridal collection
- Fancy Sarees — Party wear
- Under ₹2,000 — Budget picks

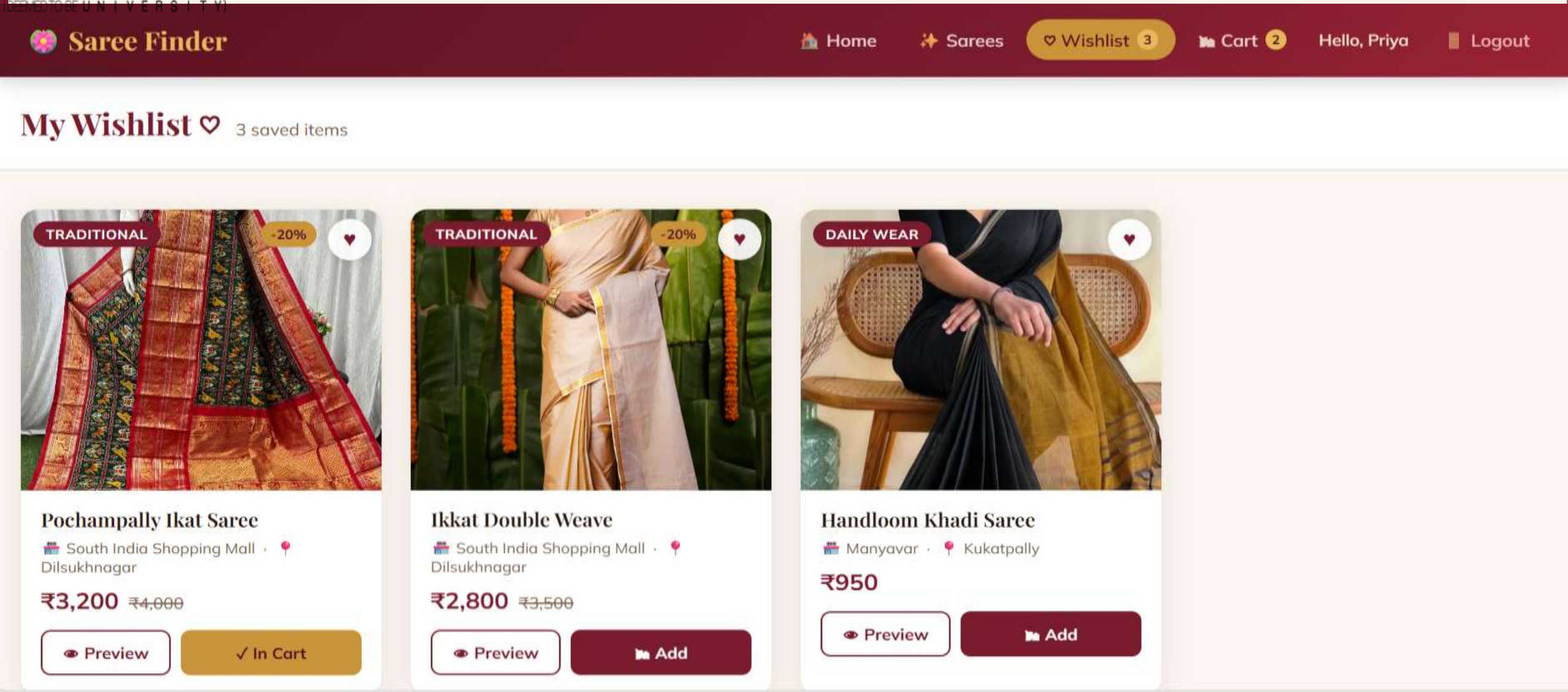
Featured Sarees

View all →

Search Page for Sarees



Wishlist Page



Cart

Saree Finder

Home | Sarees | Wishlist 3 | Cart 2 | Hello, Priya | Logout

My Cart

Pochampally Ikat Saree

South India Shopping Mall · Dilsukhnagar, Hyderabad

₹3,200

Uppada Silk Pattu

South India Shopping Mall · Dilsukhnagar, Hyderabad

₹9,800

Order Summary

Pochampally Ikat Saree	₹3,200
Uppada Silk Pattu	₹9,800
Subtotal	₹13,000
Shipping	₹150
Total	**₹13,150**

STORE LOCATIONS

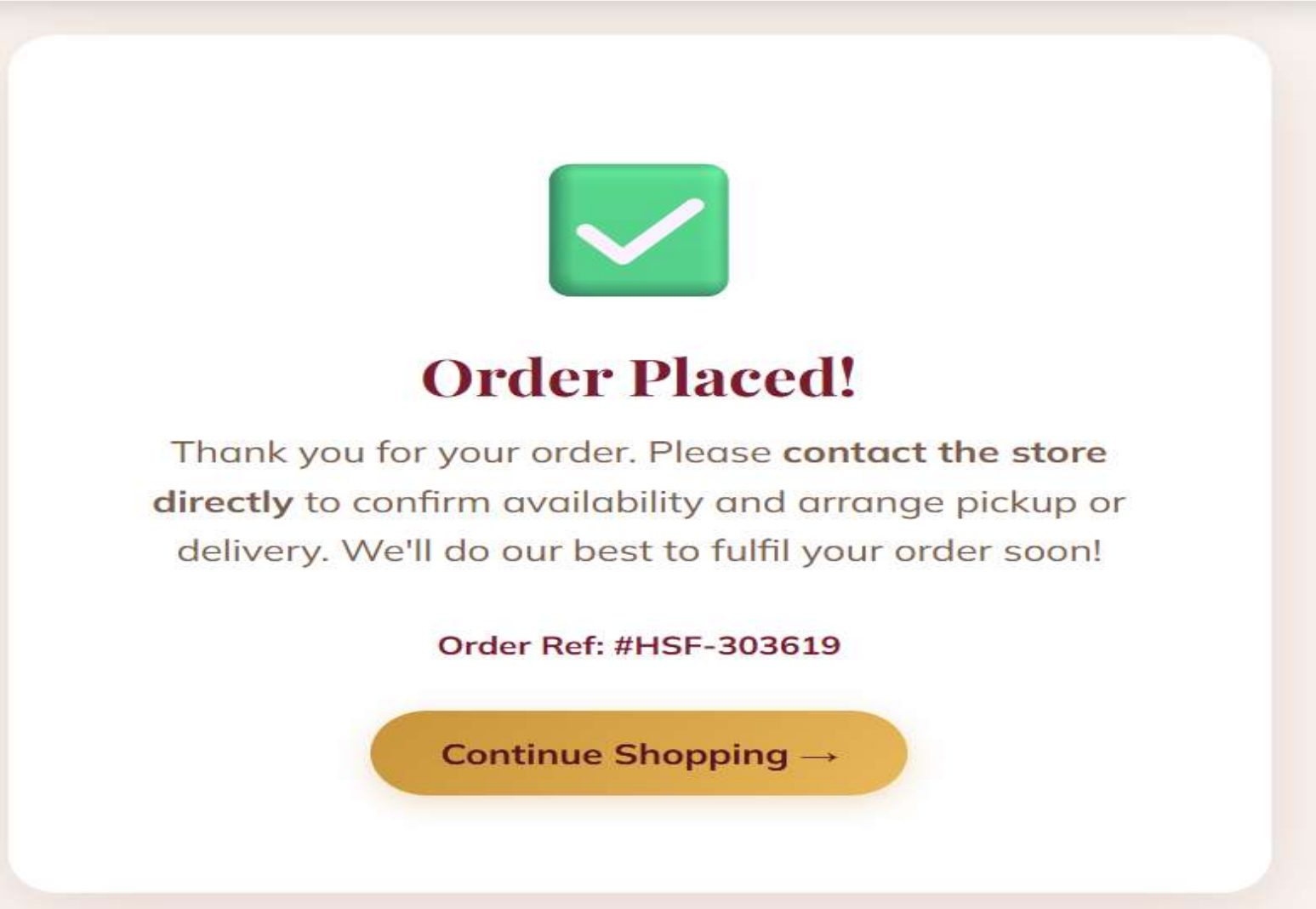
SOUTH INDIA SHOPPING MALL

— Dilsukhnagar, Hyderabad

Place Order ✓

By placing an order you agree to contact the store directly to confirm stock.

Advance Booking Process Status



Admin Page

Admin Panel

Chennai Shopping Mall | Hello, Meena | Logout

Chennai Shopping Mall

MANAGEMENT

- Inventory
- Add Saree
- Store Info

ACCOUNT

- Logout

Inventory

Managing sarees for **Chennai Shopping Mall** — data auto-saved to browser storage

4	3	1	1
TOTAL	IN STOCK	OUT OF STOCK	ON SALE

↺ Reset all data to defaults

IMAGE	NAME	CATEGORY	PRICE / DISCOUNT	STOCK	ACTIONS
	Banarasi Silk Weave	Wedding	₹12,000	✓ IN STOCK	Edit, Image, Price, Out of Stock, Delete
	Soft Silk Pattu	Wedding	₹5,200 ~~₹6,500~~ -20%	✓ IN STOCK	Edit, Image, Price, Out of Stock, Delete
	Pure Linen Comfortable	Comfortable	₹2,200	✓ IN STOCK	Edit, Image, Price, Out of Stock, Delete
	Paithani Silk Saree	Wedding	₹15,000	✗ OUT OF STOCK	Edit, Image, Price, In Stock, Delete

Co's Attainments - DSA

CO1: Analysed problem statements and evaluated algorithm efficiency using Big-O notation and time complexity concepts.

CO2: Implemented searching and sorting algorithms (Linear Search, Binary Search, Bubble Sort, Selection Sort, Insertion Sort, Merge Sort, Quick Sort) to manage saree data efficiently.

CO3: Designed and implemented Abstract Data Types using arrays and linked lists to store and manipulate saree information.

CO4: Applied stacks and queues for functionalities like wishlist management and order processing.

CO5: Implemented hash-based data structures and priority queues to enable faster data retrieval and prioritized operations.

User Login

```
C:\Users\DELL\.jdks\openjdk-25.0.2\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.2\lib\idea_rt.jar=61523" -Dfile.enc

╔═══════════════════════════════════════════════╗
  🌸 HYDERABAD SAREE FINDER SYSTEM 🌸
  Java DSA Console Application – Academic Project
╚═══════════════════════════════════════════════╝

╔═══ DEMO LOGIN CREDENTIALS ═══
CUSTOMERS:
  username: user1      password: demo123
  username: anita      password: demo123
  username: kavya      password: user123

ADMINS (all password: admin123):
  admin_kalaniketan  → Kalaniketan
  admin_chennai      → Chennai Shopping Mall
  admin_southindia   → South India Shopping Mall
  admin_sheneeds     → She Needs
  admin_manyavar     → Manyavar
  admin_mangalya     → Mangalya
```

```
═══ HYDERABAD SAREE FINDER SYSTEM ═══
 1. Customer Login
 2. Admin (Salesman) Login
 3. New Customer Registration
 4. DSA Overview (CO1-CO6)
 0. Exit
Enter choice: 1

═══ CUSTOMER LOGIN ═══
Username: user1
Password: demo123
✓ Login successful! Welcome, Priya Sharma
[DSA] Auth via Open Addressing Hash Table — O(1) average lookup

Welcome, Priya Sharma! 💖

╔═══ FEATURED DEALS — TODAY'S TOP OFFERS ═══╗
[Featured 1] Bridal Kanjivaram | Kalaniketan | ₹22000
[Featured 2] Cotton Printed Daily | Kalaniketan | ₹850
[Featured 3] Net Embroidered Fancy | Kalaniketan | ₹4200
[Featured 4] Soft Silk Pattu | Chennai Shopping Mall | ₹5200
[Featured 5] Pochampally Ikat Saree | South India Shopping Mall | ₹3200
```

Mall And Budget Range Selection

```
═══ MAIN MENU — Priya Sharma ═══
 1. Select Shopping Mall
 2. Select Budget Range
 3. Select Type of Saree
 4. View Sarees
 5. Search Sarees by Keyword
 6. Wishlist  (0 items)
 7. Cart      (0 items)
 8. Generate Bill
 9. Recently Viewed
10. Clear All Filters
 0. Logout
Enter choice: 1
═══ SELECT SHOPPING MALL ═══
 1. Kalaniketan                    (4 sarees)
 2. Chennai Shopping Mall          (4 sarees)
 3. South India Shopping Mall      (4 sarees)
 4. She Needs                      (4 sarees)
 5. Manyavar                       (4 sarees)
 6. Mangalya                       (4 sarees)
 0. Any Mall (no filter)
```

```
Select: 2
✓ Mall set to: Chennai Shopping Mall
[DSA] Hash Table filterByStore() — O(n) scan, O(n/k) average

═══ MAIN MENU — Priya Sharma ═══
[Filters: Mall=Chennai Shopping Mall ]
 1. Select Shopping Mall
 2. Select Budget Range
 3. Select Type of Saree
 4. View Sarees
 5. Search Sarees by Keyword
 6. Wishlist  (0 items)
 7. Cart      (0 items)
 8. Generate Bill
 9. Recently Viewed
10. Clear All Filters
 0. Logout
Enter choice: 2
═══ SELECT BUDGET RANGE ═══
 1. ₹1,000 - ₹3,000   (Budget-friendly)
 2. ₹3,000 - ₹5,000   (Mid-range)
 3. ₹5,000 - ₹7,000   (Premium)
```

Type Of Saree

```
 4. ₹7,000+              (Luxury)
 5. Custom range
 0. Any price
Select: 1
✓ Budget set: ₹1000 - ₹3000
[DSA] Binary search on price-sorted array — O(log n + k)

═══ MAIN MENU — Priya Sharma ═══════════════════════════
[Filters: Mall=Chennai Shopping Mall Min=₹1000 Max=₹3000]
 1. Select Shopping Mall
 2. Select Budget Range
 3. Select Type of Saree
 4. View Sarees
 5. Search Sarees by Keyword
 6. Wishlist  (0 items)
 7. Cart      (0 items)
 8. Generate Bill
 9. Recently Viewed
10. Clear All Filters
 0. Logout
Enter choice: 3
═══ SELECT TYPE OF SAREE ═══════════════════════════════
```

```
═══ SELECT TYPE OF SAREE ═══════════════════════════════
 1. Silk
 2. Cotton
 3. Designer
 4. Wedding
 5. Traditional
 0. All Types
Select: 2
✓ Type set to: Cotton

═══ MAIN MENU — Priya Sharma ═══════════════════════════
[Filters: Mall=Chennai Shopping Mall Type=Cotton Min=₹1000 Max=₹3000]
 1. Select Shopping Mall
 2. Select Budget Range
 3. Select Type of Saree
 4. View Sarees
 5. Search Sarees by Keyword
 6. Wishlist  (0 items)
 7. Cart      (0 items)
 8. Generate Bill
 9. Recently Viewed
10. Clear All Filters
```

Option to add into Cart or Wishlist

```
10. Clear All Filters
 0. Logout
Enter choice: 4

= SAREES (1 items) =
ID   Name                    Type        Store                   Price                   Stock
------------------------------------------------------------------------------------------
[ 7] Pure Linen Comfortable  | Cotton    | Chennai Shopping Mall  | ₹2200                | In Stock
------------------------------------------------------------------------------------------

What would you like to do?
 1. Add to Wishlist
 2. Add to Cart
 3. View saree details
 0. Back
Choice: 2
Enter Saree ID to add to cart: 7
✓ Added to cart: Pure Linen Comfortable | ₹2200
[DSA] DoublyLinkedList insertAtTail - O(1)
```

```
== MAIN MENU - Priya Sharma ==========================
[Filters: Mall=Chennai Shopping Mall Type=Cotton Min=₹1000 Max=₹3000]
 1. Select Shopping Mall
 2. Select Budget Range
 3. Select Type of Saree
 4. View Sarees
 5. Search Sarees by Keyword
 6. Wishlist  (0 items)
 7. Cart      (1 items)
 8. Generate Bill
 9. Recently Viewed
10. Clear All Filters
 0. Logout
Enter choice: 8

╔═══════════════════════════════════════════════════╗
║          SAREE FINDER SYSTEM - INVOICE            ╕
╞═══════════════════════════════════════════════════╡
│  Customer: user1                                  │
╘═══════════════════════════════════════════════════╛
```

Display of Bill and Recently Viewed item

```
── FIRST SAREE BILL ──────────────────────────────────────
Saree Name                    : Pure Linen Comfortable
Type                          : Cotton
Store                         : Chennai Shopping Mall
Original Price                : ₹2200
Sale Price                    : ₹2200
GST                           : 5% → ₹110
─── Subtotal ──────────────────────── : ₹2310

═══════════════════════════════════════════════════════════
COMBINED BILL SUMMARY
═══════════════════════════════════════════════════════════
Total Items                   : 1 items
Items Subtotal                : ₹2200
Total GST                     : ₹110
Shipping                      : ₹150 (free above ₹3000)
───────────────────────────────────────────────────────────
GRAND TOTAL                   : ₹2460
═══════════════════════════════════════════════════════════
Thank you for shopping at Hyderabad Saree Finder! 🌸
```

```
═══ MAIN MENU - Priya Sharma ═══════════════════════════
[Filters: Mall=Chennai Shopping Mall Type=Cotton Min=₹1000 Max=₹3000]
 1. Select Shopping Mall
 2. Select Budget Range
 3. Select Type of Saree
 4. View Sarees
 5. Search Sarees by Keyword
 6. Wishlist  (0 items)
 7. Cart      (1 items)
 8. Generate Bill
 9. Recently Viewed
10. Clear All Filters
 0. Logout
Enter choice: 9

── Recently Viewed (last 5) ──
Deque contents [front → rear]:
  1. Pure Linen Comfortable | ₹2200
[DSA] Deque sliding window - insertRear O(1), deleteFront O(1)
```

Some Sample Test Cases on Random raised questions:

```
== MAIN MENU — Priya Sharma ==============================
[Filters: Mall=Chennai Shopping Mall Type=Wedding Min=₹1000 Max=₹3000]
 1. Select Shopping Mall
 2. Select Budget Range
 3. Select Type of Saree
 4. View Sarees
 5. Search Sarees by Keyword
 6. Wishlist  (0 items)
 7. Cart      (0 items)
 8. Generate Bill
 9. Recently Viewed
10. Clear All Filters
 0. Logout
Enter choice: 4
ℹ No sarees match your filters. Try adjusting them.
```

```
[Filters: Type=Designer ]
 1. Select Shopping Mall
 2. Select Budget Range
 3. Select Type of Saree
 4. View Sarees
 5. Search Sarees by Keyword
 6. Wishlist  (0 items)
 7. Cart      (0 items)
 8. Generate Bill
 9. Recently Viewed
10. Clear All Filters
 0. Logout
Enter choice: 4

== SAREES (4 items) ==
ID   Name                        Type        Store                Price                      Stock
-------------------------------------------------------------------------------------------------
[ 4] Net Embroidered Fancy       | Designer  | Kalaniketan          | ₹4300  (was ₹5000, -14%)  | In Stock
[13] Designer Georgette          | Designer  | She Needs            | ₹4500                     | In Stock
[14] Chiffon Floral Fancy        | Designer  | She Needs            | ₹1800                     | In Stock
[24] Silk Blend Party Saree      | Designer  | Mangalya             | ₹5500                     | In Stock
```

Admin Login and his Inventory

```
═══ HYDERABAD SAREE FINDER SYSTEM ═══
 1. Customer Login
 2. Admin (Salesman) Login
 3. New Customer Registration
 4. DSA Overview (CO1-CO6)
 0. Exit
Enter choice: 2

═══ ADMIN LOGIN ═══
Username: admin_kalaniketan
Password: admin123
✓ Admin login successful! Store: Kalaniketan

Welcome, Admin Ravi Kumar | Store: Kalaniketan

═══ ADMIN PANEL — Kalaniketan ═══
 1. View My Inventory
 2. Add Saree
 3. Delete Saree
 4. Update Price
 5. Update Discount
```

```
 6. Update Stock Availability
 7. Sort Sarees
 8. View All Sarees (all stores)
 9. Show Top Deals (Max-Heap)
10. DSA Demonstration
 0. Logout
Enter choice: 1

═ MY INVENTORY - Kalaniketan (4 items) ═
ID   Name                    Type        Store                Price                    Stock
──────────────────────────────────────────────────────────────────────────────────────────
[ 1] Bridal Kanjivaram       | Wedding   | Kalaniketan        | ₹22000  (was ₹25000, -12%)  | In Stock
[ 2] Kanchipuram Silk Saree  | Silk      | Kalaniketan        | ₹8500                       | In Stock
[ 3] Cotton Printed Daily    | Cotton    | Kalaniketan        | ₹850  (was ₹1000, -15%)     | In Stock
[ 4] Net Embroidered Fancy   | Designer  | Kalaniketan        | ₹4200  (was ₹5000, -16%)    | In Stock
──────────────────────────────────────────────────────────────────────────────────────────

[DSA] Hash Table filter by store - O(n/k) average

═ ADMIN PANEL - Kalaniketan ═══
 1. View My Inventory
 2. Add Saree
```

Addition of New Saree and Deletion

```
  3. Delete Saree
  4. Update Price
  5. Update Discount
  6. Update Stock Availability
  7. Sort Sarees
  8. View All Sarees (all stores)
  9. Show Top Deals (Max-Heap)
 10. DSA Demonstration
  0. Logout
Enter choice: 2
== ADD NEW SAREE ==========================================
Banaras Silk Pattu
Saree Name    : Banaras Silk Pattu
Types: Silk | Cotton | Designer | Wedding | Traditional
Type          : Silk
Original Price (₹): 4000
Discount %    : 20
In Stock (Y/N): Y
Description   : Draped with elegance gold color
✓ Saree added: Banaras Silk Pattu (ID=25)
[DSA] Hash Table insert — O(1) average
```

```
Enter choice: 3

= MY INVENTORY - Kalaniketan (5 items) =
ID   Name                      Type        Store                Price                    Stock
-------------------------------------------------------------------------------------------------
[ 1] Bridal Kanjivaram         | Wedding   | Kalaniketan        | ₹22000  (was ₹25000, -12%)  | In Stock
[ 2] Kanchipuram Silk Saree    | Silk      | Kalaniketan        | ₹8500                       | In Stock
[ 3] Cotton Printed Daily      | Cotton    | Kalaniketan        | ₹850  (was ₹1000, -15%)     | In Stock
[ 4] Net Embroidered Fancy     | Designer  | Kalaniketan        | ₹4200  (was ₹5000, -16%)    | In Stock
[25] Banaras Silk Pattu        | Silk      | Kalaniketan        | ₹3200  (was ₹4000, -20%)    | In Stock
-------------------------------------------------------------------------------------------------
[DSA] Hash Table filter by store - O(n/k) average
Enter Saree ID to delete: 3
✓ Deleted: Cotton Printed Daily
[DSA] Hash Table delete - O(1) average; LinkedList delete - O(n)
```

Updating Price and Discount

```
Enter choice: 4

= MY INVENTORY - Kalaniketan (4 items) =
ID   Name                    Type        Store                Price                    Stock
---------------------------------------------------------------------------------------------
[ 1] Bridal Kanjivaram       | Wedding   | Kalaniketan        | ₹22000  (was ₹25000, -12%)  | In Stock
[ 2] Kanchipuram Silk Saree  | Silk      | Kalaniketan        | ₹8500                       | In Stock
[ 4] Net Embroidered Fancy   | Designer  | Kalaniketan        | ₹4200  (was ₹5000, -16%)    | In Stock
[25] Banaras Silk Pattu      | Silk      | Kalaniketan        | ₹3200  (was ₹4000, -20%)    | In Stock
---------------------------------------------------------------------------------------------
[DSA] Hash Table filter by store - O(n/k) average
Enter Saree ID: 2
New original price (₹): 9000
✓ Price updated: Kanchipuram Silk Saree → ₹9000
[DSA] Hash Table search + update - O(1) average
```

```
Enter choice: 5

= MY INVENTORY - Kalaniketan (4 items) =
ID   Name                    Type        Store                Price                    Stock
---------------------------------------------------------------------------------------------
[ 1] Bridal Kanjivaram       | Wedding   | Kalaniketan        | ₹22000  (was ₹25000, -12%)  | In Stock
[ 2] Kanchipuram Silk Saree  | Silk      | Kalaniketan        | ₹9000                       | In Stock
[ 4] Net Embroidered Fancy   | Designer  | Kalaniketan        | ₹4200  (was ₹5000, -16%)    | In Stock
[25] Banaras Silk Pattu      | Silk      | Kalaniketan        | ₹3200  (was ₹4000, -20%)    | In Stock
---------------------------------------------------------------------------------------------
[DSA] Hash Table filter by store - O(n/k) average
Enter Saree ID: 4
New discount % (0-90): 14
✓ Discount updated: Net Embroidered Fancy → 14%
[DSA] Hash Table search + update - O(1) average
```

Stock Availability and Sorting Order

```
Enter choice: 6

= MY INVENTORY - Kalaniketan (4 items) =
ID   Name                    Type        Store                Price                    Stock
------------------------------------------------------------------------------------------
[ 1] Bridal Kanjivaram       | Wedding   | Kalaniketan        | ₹22000  (was ₹25000, -12%)  | In Stock
[ 2] Kanchipuram Silk Saree  | Silk      | Kalaniketan        | ₹9000                       | In Stock
[ 4] Net Embroidered Fancy   | Designer  | Kalaniketan        | ₹4300  (was ₹5000, -14%)    | In Stock
[25] Banaras Silk Pattu      | Silk      | Kalaniketan        | ₹3200  (was ₹4000, -20%)    | In Stock
------------------------------------------------------------------------------------------
[DSA] Hash Table filter by store - O(n/k) average
Enter Saree ID:
4
In Stock (Y/N): N
✓ Stock updated: Kanchipuram Silk Saree → Out of Stock
[DSA] Hash Table search + update - O(1) average
```

```
Enter choice: 7
= SORT SAREES ==================================
Sort by:
 1. Price (Low → High)
 2. Price (High → Low)
 3. Discount (Highest first)
 4. Name (A → Z)
Choose field: 1

Algorithm:
 1. Bubble Sort   - O(n²)       best for tiny n
 2. Selection Sort- O(n²)       minimizes swaps
 3. Insertion Sort- O(n) best   best for nearly-sorted
 4. Merge Sort    - O(n log n)  stable, guaranteed
 5. Quick Sort    - O(n log n)  fastest in practice
Choose algorithm: 3

= SORTED BY PRICE_ASC using INSERTION SORT (24 items) =
ID   Name                    Type        Store                Price                    Stock
------------------------------------------------------------------------------------------
[19] Handloom Khadi Saree    | Cotton    | Manyavar           | ₹950                     | In Stock
[11] Mangalagiri Cotton      | Cotton    | South India Shopping Mall | ₹1100                | In Stock
```

[11] Mangalagiri Cotton	Cotton	South India Shopping Mall	₹1180	In Stock
[22] Zari Border Cotton	Cotton	Mangalya	₹1480	In Stock
[28] Modal Silk Comfortable	Silk	Manyavar	₹1650	In Stock
[14] Chiffon Floral Fancy	Designer	She Needs	₹1880	In Stock
[7] Pure Linen Comfortable	Cotton	Chennai Shopping Mall	₹2280	In Stock
[23] Kalamkari Printed	Traditional	Mangalya	₹2610 (was ₹3000, -13%)	In Stock
[12] Ikkat Double Weave	Traditional	South India Shopping Mall	₹2800 (was ₹3500, -20%)	In Stock
[9] Pochampally Ikat Saree	Traditional	South India Shopping Mall	₹3200 (was ₹4000, -20%)	In Stock
[25] Banaras Silk Pattu	Silk	Kalaniketan	₹3200 (was ₹4000, -20%)	In Stock
[15] Chanderi Silk Cotton	Silk	She Needs	₹3880	In Stock
[4] Net Embroidered Fancy	Designer	Kalaniketan	₹4300 (was ₹5000, -14%)	In Stock
[13] Designer Georgette	Designer	She Needs	₹4500	In Stock
[18] Tussar Silk Natural	Silk	Manyavar	₹4800	In Stock
[6] Soft Silk Pattu	Silk	Chennai Shopping Mall	₹5200 (was ₹6500, -20%)	In Stock
[24] Silk Blend Party Saree	Designer	Mangalya	₹5500	In Stock
[16] Organza Silk Fancy	Designer	She Needs	₹6200	OUT OF STOCK
[17] Gadwal Silk Saree	Traditional	Manyavar	₹6500	In Stock
[2] Kanchipuram Silk Saree	Silk	Kalaniketan	₹9000	OUT OF STOCK
[10] Uppada Silk Pattu	Silk	South India Shopping Mall	₹9790 (was ₹11000, -11%)	In Stock
[5] Banarasi Silk Weave	Silk	Chennai Shopping Mall	₹12000	In Stock
[8] Paithani Silk Saree	Wedding	Chennai Shopping Mall	₹15000	OUT OF STOCK
[21] Bridal Silk Lehenga	Wedding	Mangalya	₹18000 (was ₹20000, -10%)	In Stock
[1] Bridal Kanjivaram	Wedding	Kalaniketan	₹22000 (was ₹25000, -12%)	In Stock

SORTING ALGORITHM COMPLEXITY (CO1)

Algorithm	Best Ω	Average Θ	Worst O	Space	Stable
Bubble Sort	$\Omega(n)$	$\Theta(n^2)$	$O(n^2)$	$O(1)$	Yes
Selection Sort	$\Omega(n^2)$	$\Theta(n^2)$	$O(n^2)$	$O(1)$	No
Insertion Sort	$\Omega(n)$	$\Theta(n^2)$	$O(n^2)$	$O(1)$	Yes
Merge Sort	$\Omega(n \log n)$	$\Theta(n \log n)$	$O(n \log n)$	$O(n)$	Yes
Quick Sort	$\Omega(n \log n)$	$\Theta(n \log n)$	$O(n^2)$	$O(\log n)$	No

Display of All Sarees and Top 5 discounted and popular

```
Enter choice: 8

= ALL SAREES - All Stores (24 items) =
ID   Name                        Type         Store                    Price                    Stock
------------------------------------------------------------------------------------------------------
[ 1] Bridal Kanjivaram           | Wedding    | Kalaniketan            | ₹22000  (was ₹25000, -12%)  | In Stock
[ 2] Kanchipuram Silk Saree      | Silk       | Kalaniketan            | ₹9000                       | OUT OF STOCK
[ 4] Net Embroidered Fancy       | Designer   | Kalaniketan            | ₹4300  (was ₹5000, -14%)    | In Stock
[ 5] Banarasi Silk Weave         | Silk       | Chennai Shopping Mall  | ₹12000                      | In Stock
[ 6] Soft Silk Pattu             | Silk       | Chennai Shopping Mall  | ₹5200  (was ₹6500, -20%)    | In Stock
[ 7] Pure Linen Comfortable      | Cotton     | Chennai Shopping Mall  | ₹2200                       | In Stock
[ 8] Paithani Silk Saree         | Wedding    | Chennai Shopping Mall  | ₹15000                      | OUT OF STOCK
[ 9] Pochampally Ikat Saree      | Traditional | South India Shopping Mall | ₹3200  (was ₹4000, -20%)  | In Stock
[10] Uppada Silk Pattu           | Silk       | South India Shopping Mall | ₹9790  (was ₹11000, -11%) | In Stock
[11] Mangalagiri Cotton          | Cotton     | South India Shopping Mall | ₹1100                     | In Stock
[12] Ikkat Double Weave          | Traditional | South India Shopping Mall | ₹2800  (was ₹3500, -20%)  | In Stock
[13] Designer Georgette          | Designer   | She Needs              | ₹4500                       | In Stock
[14] Chiffon Floral Fancy        | Designer   | She Needs              | ₹1800                       | In Stock
[15] Chanderi Silk Cotton        | Silk       | She Needs              | ₹3800                       | In Stock
[16] Organza Silk Fancy          | Designer   | She Needs              | ₹6200                       | OUT OF STOCK
[17] Gadwal Silk Saree           | Traditional | Manyavar               | ₹6500                       | In Stock
[18] Tussar Silk Natural         | Silk       | Manyavar               | ₹4800                       | In Stock
```

```
Enter choice: 9

╔═ TOP 5 DISCOUNTED SAREES (Max-Heap) ═══════════╗
│ 1. Soft Silk Pattu          | -20% | ₹5200    │
│ 2. Pochampally Ikat Saree   | -20% | ₹3200    │
│ 3. Ikkat Double Weave       | -20% | ₹2800    │
│ 4. Banaras Silk Pattu       | -20% | ₹3200    │
│ 5. Cotton Printed Daily     | -15% | ₹850     │
╚═══════════════════════════════════════════════╝

╔═ TOP 5 POPULAR SAREES (Priority Queue) ═══╗
│ 1. Banarasi Silk Weave      | Score:90  | ₹12000  │
│ 2. Bridal Silk Lehenga      | Score:88  | ₹18000  │
│ 3. Bridal Kanjivaram        | Score:85  | ₹22000  │
│ 4. Designer Georgette       | Score:82  | ₹4500   │
│ 5. Paithani Silk Saree      | Score:80  | ₹15000  │
╚═══════════════════════════════════════════╝

[DSA] Max-Heap extractMax - O(log n); Priority Queue dequeue - O(log n)
```

DSA Demonstration

```
Enter choice: 10
═══ DSA DEMONSTRATION (CO1-CO6) ═══

[CO1] Search & Sort Complexity:

═══ SEARCH ALGORITHM COMPARISON (CO1) ═══
```

n (array size)	Linear Search	Binary Search
n = 10	10	4
n = 100	100	7
n = 1000	1000	10
n = 10000	10000	14
n = 100000	100000	17

```
═══ SORTING ALGORITHM COMPLEXITY (CO1) ═══
```

Algorithm	Best Ω	Average Θ	Worst O	Space	Stable
Bubble Sort	Ω(n)	Θ(n^2)	O(n^2)	O(1)	Yes
Selection Sort	Ω(n^2)	Θ(n^2)	O(n^2)	O(1)	No
Insertion Sort	Ω(n)	Θ(n^2)	O(n^2)	O(1)	Yes
Merge Sort	Ω(n log n)	Θ(n log n)	O(n log n)	O(n)	Yes
Quick Sort	Ω(n log n)	Θ(n log n)	O(n^2)	O(logn)	No

```
[CO2] Hash Table Structure:

Hash Table Structure (Chaining) - tableSize=31
Bucket[ 1]: [ID:1 Bridal Kanjivaram] → null
Bucket[ 2]: [ID:2 Kanchipuram Silk Saree] → null
Bucket[ 4]: [ID:4 Net Embroidered Fancy] → null
Bucket[ 5]: [ID:5 Banarasi Silk Weave] → null
Bucket[ 6]: [ID:6 Soft Silk Pattu] → null
Bucket[ 7]: [ID:7 Pure Linen Comfortable] → null
Bucket[ 8]: [ID:8 Paithani Silk Saree] → null
Bucket[ 9]: [ID:9 Pochampally Ikat Saree] → null
Bucket[10]: [ID:10 Uppada Silk Pattu] → null
Bucket[11]: [ID:11 Mangalagiri Cotton] → null
Bucket[12]: [ID:12 Ikkat Double Weave] → null
Bucket[13]: [ID:13 Designer Georgette] → null
Bucket[14]: [ID:14 Chiffon Floral Fancy] → null
Bucket[15]: [ID:15 Chanderi Silk Cotton] → null
Bucket[16]: [ID:16 Organza Silk Fancy] → null
```

```
Bucket[13]: [ID:13 Designer Georgette] → null
Bucket[14]: [ID:14 Chiffon Floral Fancy] → null
Bucket[15]: [ID:15 Chanderi Silk Cotton] → null
Bucket[16]: [ID:16 Organza Silk Fancy] → null
Bucket[17]: [ID:17 Gadwal Silk Saree] → null
Bucket[18]: [ID:18 Tussar Silk Natural] → null
Bucket[19]: [ID:19 Handloom Khadi Saree] → null
Bucket[20]: [ID:20 Modal Silk Comfortable] → null
Bucket[21]: [ID:21 Bridal Silk Lehenga] → null
Bucket[22]: [ID:22 Zari Border Cotton] → null
Bucket[23]: [ID:23 Kalamkari Printed] → null
Bucket[24]: [ID:24 Silk Blend Party Saree] → null
Bucket[25]: [ID:25 Banaras Silk Pattu] → null
Load factor: 0.77 (24 entries / 31 buckets)

[CO2] Cycle detection on catalogue list:
Cycle detected in catalogue list: false (expected: false)

[CO3] Top Discounted Sarees (Max-Heap):

╔═══ TOP 3 DISCOUNTED SAREES (Max-Heap) ═══════════╗
1. Soft Silk Pattu              | -20% | ₹5200
```

```
[CO3] Top Discounted Sarees (Max-Heap):

╔═══ TOP 3 DISCOUNTED SAREES (Max-Heap) ═══════════╗
1. Soft Silk Pattu              | -20% | ₹5200
2. Pochampally Ikat Saree       | -20% | ₹3200
3. Ikkat Double Weave           | -20% | ₹2800
╚═══════════════════════════════════════════════════╝

[CO3] Top Popular Sarees (Priority Queue):

╔═══ TOP 3 POPULAR SAREES (Priority Queue) ═══╗
1. Banarasi Silk Weave          | Score:90  | ₹12000
2. Bridal Silk Lehenga          | Score:88  | ₹18000
3. Bridal Kanjivaram            | Score:85  | ₹22000
╚═══════════════════════════════════════════════╝
```

New Customer Registration

```
10. DSA Demonstration
 0. Logout
Enter choice: 0
Admin logout. Goodbye, Ravi Kumar!

== HYDERABAD SAREE FINDER SYSTEM ==========================
 1. Customer Login
 2. Admin (Salesman) Login
 3. New Customer Registration
 4. DSA Overview (CO1-CO6)
 0. Exit
Enter choice: 3

== NEW CUSTOMER REGISTRATION ==============================
Choose username   : priya@gmail.com
Password          : priya23
Your full name    : Priyanka
Phone number      : 7777777789
✓ Account created! You can now login as: priya@gmail.com

== HYDERABAD SAREE FINDER SYSTEM ==========================
```

Conclusion

- Developed a working saree browsing and management system using HTML, CSS, JavaScript and Data Structures concepts.
- Customers can browse, search, save to wish-list, and add sarees to cart from multiple Hyderabad stores.
- Admins can manage their store sarees by adding, editing, deleting items, and updating stock details.
- Data is stored locally using browser local Storage and structured data handling without requiring a server or database.
- Identified and fixed an issue where multiple admins could not view their store sarees correctly.
- The system works smoothly on desktop browsers and demonstrates practical use of DSA for data management.

Geo Tag Photos

GPS Map Camera
Hyderabad, Telangana, India
Gcw3+wm6, Aleap Industrial Area, Gajularamaram,
Hyderabad, Telangana 500043, India
Lat 17.547381° Long 78.404655°
Saturday, 07/03/2026 02:32 PM GMT +05:30
Google

References

•MDN Web Docs – HTML5, CSS3, and JavaScript Documentation: https://developer.mozilla.org

•W3Schools – HTML, CSS, JavaScript and Web Development Tutorials: https://www.w3schools.com

•JavaScript.info – JavaScript Learning Guide: https://javascript.info

•GeeksforGeeks – Data Structures and Algorithms Concepts: https://www.geeksforgeeks.org

•Programiz – Data Structures and Algorithms Tutorials: https://www.programiz.com

•Project README – PreSareeFinder Documentation (Internal Reference)

THANK YOU